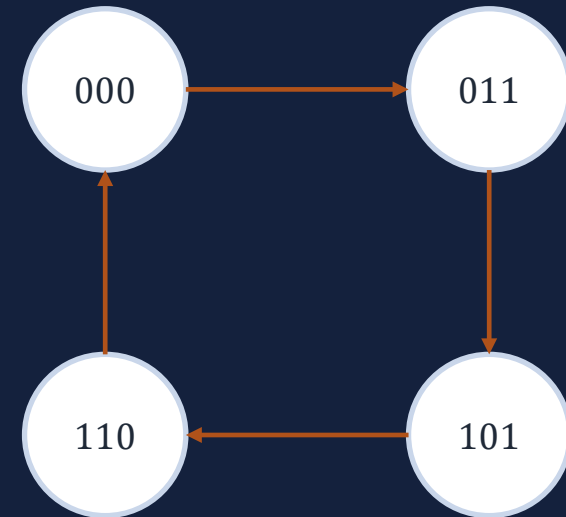


Lecture 29

Comprehensive Final Review

One worked problem per unit · numbers to memory ·
sequential design and unused states · mixed practice set · exam
strategy

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. —
Chapters 1–7 (comprehensive review)



Today's roadmap

1

§3.3

Foundations

Numbers, algebra, K-maps

2

§4.11

Combinational

Blocks and design

3

§5.8

Sequential

Design, counters

4

§7.3

Memory

Expansion, Hamming

By the end of class you can...

- Do signed arithmetic and base conversion quickly
- Minimize with don't-cares and implement on a MUX
- Design a counter and fix an unused-state lockout
- Trace shift-register counters
- Expand memories and correct single-bit errors

Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; how the final is organised	—
0:05 – 0:25	Numbers, K-maps	Reviews 1, 2
0:25 – 0:40	Multiplexer implementation	Review 3
0:40 – 0:50	Break	
0:50 – 1:20	Sequential design; counters	Reviews 4, 5
1:20 – 1:30	Expansion and Hamming	Reviews 6, 7
1:30 – 1:50	Mixed practice set; strategy	Practice set

Review 1 — Signed arithmetic

§1.6

REVIEW

Compute $(-37) + (-58)$ in 8-bit 2's complement. Overflow? Then convert $(3B)_{16}$ to binary and decimal.

-37	1	1	0	1	1	0	1	1
-58	1	1	0	0	0	1	1	0
sum	1	1	0	1	0	0	0	1

The end carry (highlighted) is dropped.

Carry into and out of the sign bit are both 1: $V = 0$.

- 1 $37 = 00100101$; invert 11011010 , add 1: $-37 = \mathbf{11011011}$
- 2 $58 = 00111010$; invert 11000101 , add 1: $-58 = \mathbf{11000110}$
- 3 Add: $1\ 10100001$; drop the end carry
- 4 Sign 1: negative. Invert 01011110 , add 1: $01011111 = 95$
- 5 So the sum is $\mathbf{-95}$ ✓
- 6 Carries into and out of the sign are both 1: $V = 0$
- 7 $(3B)_{16}$: $3 = 0011$, $B = 1011 \rightarrow \mathbf{0011\ 1011}$
- 8 $3 \cdot 16 + 11 = \mathbf{59}$

Review 2 — K-map with don't-cares

§3.5

REVIEW

Minimize $F = \Sigma m(0,2,3,4,8,10,11,15) + d(6,12)$.

AB \ CD		CD			
		00	01	11	10
00	m0	1		1	1
	m1				
01	m4	1			
	m5				X
11	m12	X		1	
	m13				
10	m8	1		1	1
	m9				

- 1 Place 1s at 0, 2, 3, 4, 8, 10, 11, 15; X at 6, 12
- 2 Column $CD = 00$: m0, m4, m8 and X at m12 $\rightarrow C'D'$
- 3 Quad m2, m3, m10, m11: $B = 0, C = 1 \rightarrow B'C$
- 4 m15 is left: pair with m11 (1011, 1111) $\rightarrow ACD$
- 5 X at m6 would not enlarge a needed group: leave it 0
- 6 Every 1 is covered: $F = ACD + C'D' + B'C$

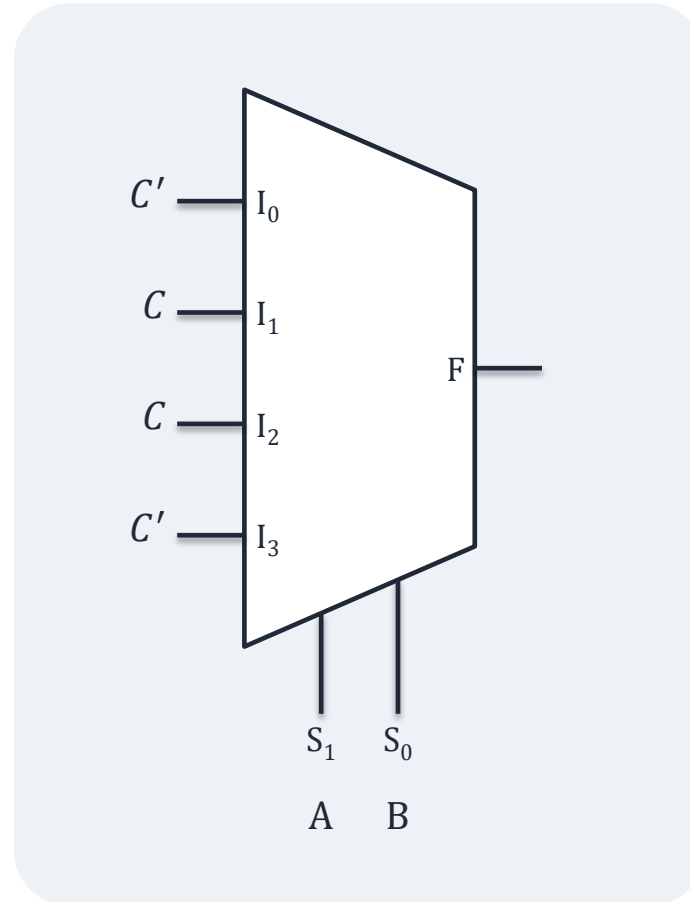
Review 3 — Function on a MUX

§4.11

REVIEW

Implement $F(A, B, C) = \Sigma m(0, 3, 5, 6)$ on a 4-to-1 MUX with A, B on the selects. What function is it?

A B	C=0	C=1	Input
00	1	0	$I_0 = C'$
01	0	1	$I_1 = C$
10	0	1	$I_2 = C$
11	1	0	$I_3 = C'$



- 1 For each AB : F at $C = 0$, then $C = 1$
- 2 $AB = 00$: $m_0 = 1, m_1 = 0 \rightarrow C'$
- 3 $AB = 01$: $m_2 = 0, m_3 = 1 \rightarrow C$
- 4 $AB = 10$: $m_4 = 0, m_5 = 1 \rightarrow C$
- 5 $AB = 11$: $m_6 = 1, m_7 = 0 \rightarrow C'$
- 6 0, 3, 5, 6 each have an even number of 1s
- 7 $F = (A \oplus B \oplus C)'$: even parity

Break

10 minutes — we resume at the 0:50 mark

Next: sequential design, counters and memory

Review 4 — Design a counter

§5.8

REVIEW

Design a counter that repeats $000 \rightarrow 011 \rightarrow 101 \rightarrow 110$ with D flip-flops. Use the unused states as don't-cares.

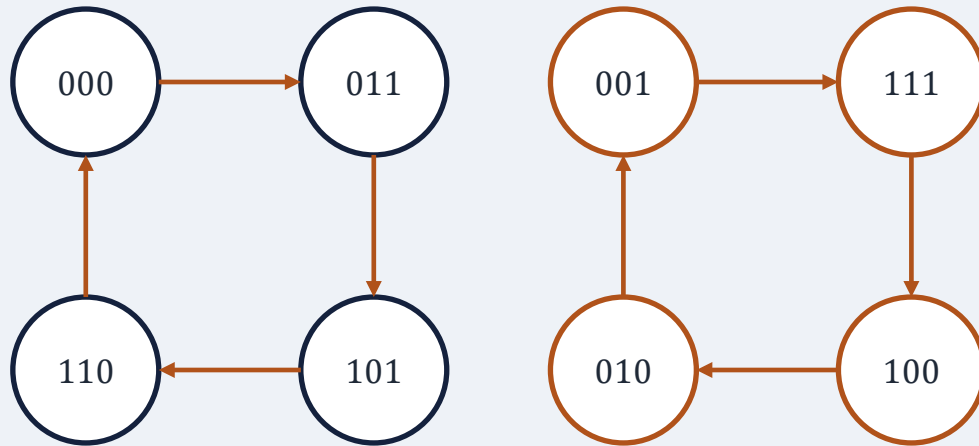
ABC	next	DA DB DC
000	011	0 1 1
011	101	1 0 1
101	110	1 1 0
110	000	0 0 0
001	X	X X X
010	X	X X X
100	X	X X X
111	X	X X X

- 1 Four states, three flip-flops; unused 001, 010, 100, 111 $\rightarrow$ X
- 2 D flip-flops: $D = \text{next-state bit}$
- 3 Next $A = 0, 1, 1, 0$ for 000, 011, 101, 110: equals $C \rightarrow D_A = C$
- 4 Next $B = 1, 0, 1, 0$: the opposite of $B \rightarrow D_B = B'$
- 5 Next $C = 1, 1, 0, 0$: the opposite of $A \rightarrow D_C = A'$
- 6 No gates at all besides one inverter per bit
- 7 Next: check where the unused states go

Review 4 — Unused states lock out

§5.8

With $D_C = A'$: a second cycle



Left: the count. Right: $001 \rightarrow 111 \rightarrow 100 \rightarrow 010 \rightarrow 001$, which never returns.

e.g. 001 : $D_A = C = 1$, $D_B = B' = 1$, $D_C = A' = 1 \rightarrow 111$

Power up in 001 and the counter is stuck forever.

The fix

Choose the don't-cares so every unused state leads into the count.

$D_C = A' + BC$ (one more AND gate) sends

$001 \rightarrow 111 \rightarrow 101$, $010 \rightarrow 001$, $100 \rightarrow 010$

e.g. 111 : $D_A = 1$, $D_B = 0$, $D_C = 0 + 1 = 1 \rightarrow 101$

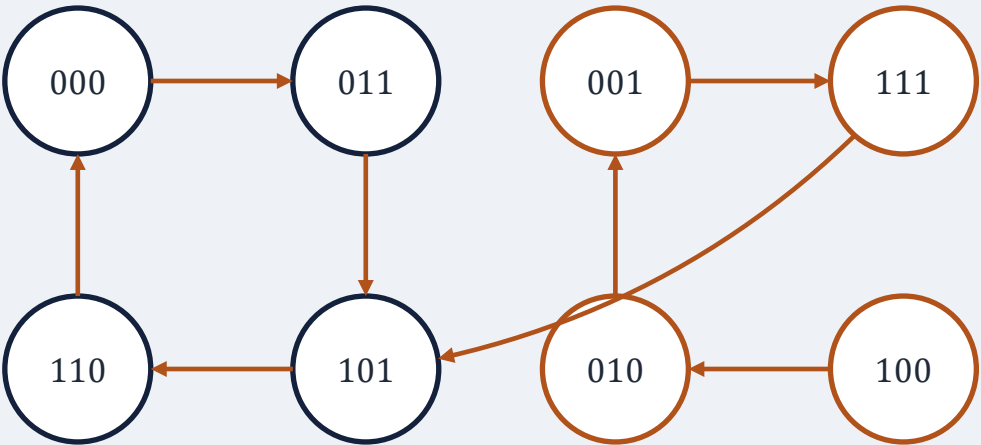
The count is unchanged: $BC = 0$ on 101 and 110 , $A' = 1$ on 000 and 011

Every unused state now reaches the count within 4 clocks: self-correcting.

Found by trying every assignment of the unused next states and keeping the cheapest that recovers.

Review 4 — The corrected counter

With $D_C = A' + BC$: every state recovers



Unused	next	path to the count
100	010	100 → 010 → 001 → 111 → 101
010	001	010 → 001 → 111 → 101
001	111	001 → 111 → 101
111	101	111 → 101

- 1 Follow each unused state with the new equations
- 2 100 → 010 → 001 → 111 → 101: longest, **4 clocks**
- 3 Cost: one AND gate and one OR gate

Review 5 — A Johnson counter

§6.5

REVIEW

A 3-bit Johnson counter starts at 000. List its states and decode state 110.

Clock	$Q_2Q_1Q_0$
0	000
1	100
2	110
3	111
4	011
5	001

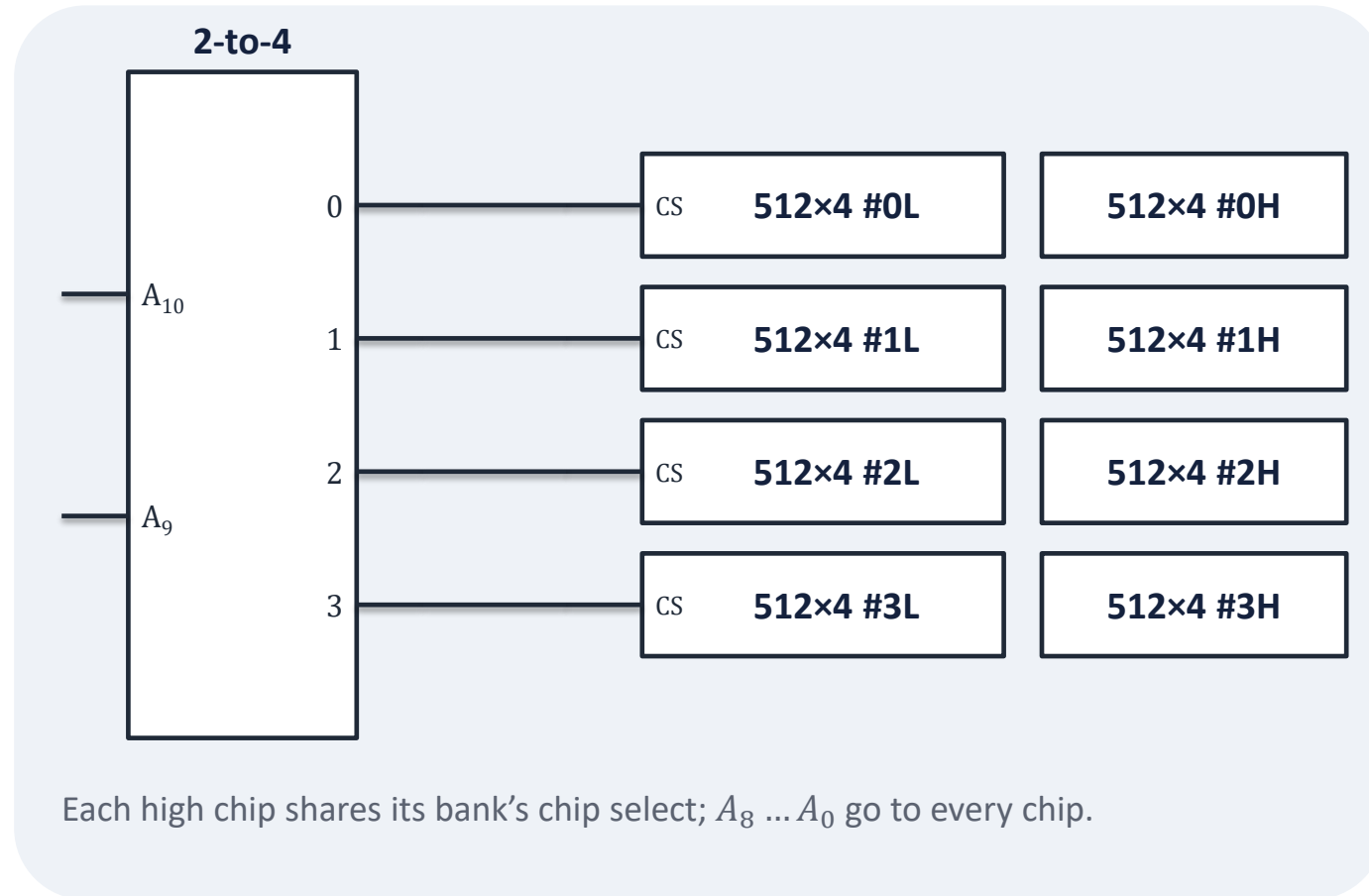
- 1 Shift right each clock; Q_0' feeds back into Q_2
- 2 States: $000 \rightarrow 100 \rightarrow 110 \rightarrow 111 \rightarrow 011 \rightarrow 001$
- 3 Six states ($2n$ for $n = 3$), then back to 000
- 4 Each state has a unique pair of adjacent bits
- 5 110 is the only state with $Q_1 = 1, Q_0 = 0$
- 6 Decode 110 with Q_1Q_0' (one 2-input AND)

Review 6 — Memory expansion

§7.3

REVIEW

Build $2K \times 8$ from 512×4 chips. Which bank holds address $0x4A0$?



- 1 Width: $8/4 = 2$ chips per bank
- 2 Depth: $2048/512 = 4$ banks → **8 chips**
- 3 $2K = 2^{11}$: 11 lines; $512 = 2^9$: $A_8 - A_0$ to every chip
- 4 $A_{10}A_9 \rightarrow$ the 2-to-4 decoder picks the bank
- 5 $4A0_{16} = 100\ 1010\ 0000$ (11 bits)
- 6 $A_{10}A_9 = 10 \rightarrow$ **bank 2**

Review 7 — Hamming (7,4)

§7.4

REVIEW

Encode 1001. The word 0011101 is received: correct it.

	1	2	3	4	5	6	7
sent	0	0	1	1	0	0	1
received	0	0	1	1	1	0	1

- 1 Data 1, 0, 0, 1 at positions 3, 5, 6, 7
- 2 $P_1 = 1 \oplus 0 \oplus 1 = 0$; $P_2 = 1 \oplus 0 \oplus 1 = 0$
- 3 $P_4 = 0 \oplus 0 \oplus 1 = 1 \rightarrow \mathbf{0011001}$
- 4 Received 0011101: 1s at 3, 4, 5, 7
- 5 XOR: $011 \oplus 100 \oplus 101 \oplus 111 = 101 = 5$: bit 5 is wrong
- 6 Flip it $\rightarrow \mathbf{0011001} \checkmark$

Mixed practice set — 15 minutes

§2.4

YOUR TURN

≈ 15 min

1. Multiply 1011×0110 in binary. Check in decimal.
2. Minimize $F(A, B, C) = \Sigma m(1, 3, 5, 7)$.
3. $t_{pd} = 5$, $t_{comb} = 12$, $t_{setup} = 3$ ns: f_{max} ?
4. What does a JK flip-flop do with $J = K = 1$?
5. How many chips of $1K \times 4$ make $4K \times 8$?
6. Which PLD shares product terms between outputs?

Work alone.

Mixed practice set — Solutions

§2.4

SOLUTION

1. 1011×0110 : multiplier bits 1 and 2 are 1
2. Partial products 10110 and 101100; add: **1000010**
3. $66 = 11 \times 6$ ✓
4. 1, 3, 5, 7 are every row with $C = 1$: a quad $\rightarrow F = C$
5. $T \geq 5 + 12 + 3 = 20 \text{ ns} \rightarrow$ **50 MHz**
6. $J = K = 1$: it **toggles** ($Q^+ = Q'$)
7. Width $8/4 = 2$, depth $4K/1K = 4 \rightarrow$ **8 chips**
8. The **PLA**: its OR array is programmable,
9. so one product term can feed several outputs

Final exam strategy

During the exam

Skim everything first; start with what you know.

Name the procedure and number the steps: partial credit follows structure.

State every assumption: MSB, state assignment, active levels.

Check each answer on one row or by converting back.

Bring

Pencil, eraser, calculator.

Know by heart

DeMorgan, consensus, $x + x'y = x + y$

Gray-code map labels; the full adder

$Q^+ = D, JQ' + K'Q, T \oplus Q$ and the excitation tables

$T \geq t_{pd} + t_{comb} + t_{setup}$

$2^k \times n$ memories; $2^k \geq n + k + 1$

The Key takeaways slide of every lecture is the checklist.

Key takeaways

Numbers

2's complement, $V = C_{in} \oplus C_{out}$

Minimize

K-map, tabular, don't-cares

Combinational

decoders, MUXes, adders

Sequential

diagram $\rightarrow$ table $\rightarrow$ maps $\rightarrow$ check

Counters

check unused states

Memory

$2^k \times n$, Hamming

Every digital system is registers joined by combinational logic.

Next: the final exam — good luck